

ACIS-X Test Results

Generated: 2026-07-23

Python: 3.14.6 · **pytest:** 9.1.1 · **Platform:** macOS (darwin, arm64)

Commands run: `python -m pytest tests/ -v` and `python -m pytest tests/suite/ -v`

Summary

Command	Scope	Collected	Passed	Failed	Duration
<code>python -m pytest tests/ -v</code>	Full suite (regular + suite)	162	162	0	46.86 s
<code>python -m pytest tests/suite/ -v</code>	Integration/suite only	45	45	0	33.23 s

Overall result:  All tests pass. Zero failures.

Command 1 — `python -m pytest tests/ -v`

Environment

```
platform darwin -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0
rootdir: /Users/johangeorge/ACIS-X
configfile: pytest.ini
plugins: anyio-4.14.2
collected 162 items
```

Results by File

`tests/regular_tests/` — 117 tests

File	Tests	Result
<code>test_aggregator_null_safety.py</code>	5	 All passed
<code>test_canonical_consumer_group_scaling.py</code>	4	 All passed
<code>test_enrichment_ablation.py</code>	2	 All passed
<code>test_exactly_once_processing.py</code>	2	 All passed
<code>test_integration_pipeline.py</code>	6	 All passed
<code>test_ml_payment_prediction.py</code>	3	 All passed
<code>test_overpayment_clamping.py</code>	2	 All passed
<code>test_query_client_thread_isolation.py</code>	3	 All passed
<code>test_recovery_action_correctness.py</code>	11	 All passed
<code>test_recovery_storm_prevention.py</code>	1	 All passed
<code>test_risk_score_monotonicity.py</code>	4	 All passed
<code>test_self_healing_concurrent.py</code>	1	 All passed
<code>test_throughput_scaling.py</code>	1	 All passed
<code>test_unit_architecture_fixes.py</code>	19	 All passed
<code>test_unit_circuit_breaker.py</code>	4	 All passed
<code>test_unit_datetime_tz.py</code>	7	 All passed
<code>test_unit_event_validation.py</code>	4	 All passed

File	Tests	Result
test_unit_external_agent.py	4	✓ All passed
test_unit_kafka_client.py	7	✓ All passed
test_unit_placement.py	3	✓ All passed
test_unit_query_client.py	4	✓ All passed
test_unit_schema.py	4	✓ All passed
test_unit_self_healing.py	5	✓ All passed
test_unit_self_healing_health_routing.py	8	✓ All passed

tests/suite/ — 45 tests

File	Tests	Result
test_dynamic_discovery.py	2	✓ All passed
test_integration_contracts.py	6	✓ All passed
test_intelligence_comparison.py	3	✓ All passed
test_lifecycle_trace.py	4	✓ All passed
test_performance.py	3	✓ All passed
test_self_healing_proof.py	5	✓ All passed
test_unit_core.py	22	✓ All passed

Individual Test List (Regular Tests)

► Click to expand all 117 regular test names

```
tests/regular_tests/test_aggregator_null_safety.py::TestAggregatorNullSafety::test_null_safe_risk_fusion[financial=0.6_litigation=0.3] PASSED
tests/regular_tests/test_aggregator_null_safety.py::TestAggregatorNullSafety::test_null_safe_risk_fusion[financial=None_litigation=0.5] PASSED
tests/regular_tests/test_aggregator_null_safety.py::TestAggregatorNullSafety::test_null_safe_risk_fusion[financial=0.6_litigation=None] PASSED
tests/regular_tests/test_aggregator_null_safety.py::TestAggregatorNullSafety::test_null_safe_risk_fusion[financial=None_litigation=None] PASSED
tests/regular_tests/test_aggregator_null_safety.py::TestAggregatorNullSafety::test_none_financial_does_not_overwrite_valid_score PASSED
tests/regular_tests/test_canonical_consumer_group_scaling.py::TestCanonicalConsumerGroupScaling::test_same_group_id_load_balances PASSED
tests/regular_tests/test_canonical_consumer_group_scaling.py::TestCanonicalConsumerGroupScaling::test_unique_group_ids_broadcast PASSED
tests/regular_tests/test_canonical_consumer_group_scaling.py::TestCanonicalConsumerGroupScaling::test_acis_agents_use_canonical_group_ids PASSED
tests/regular_tests/test_canonical_consumer_group_scaling.py::TestCanonicalConsumerGroupScaling::test_multiple_replicas_share_canonical_group_id PASSED
tests/regular_tests/test_enrichment_ablation.py::TestEnrichmentAblation::test_enrichment_reduces_variance_for_public_companies PASSED
tests/regular_tests/test_enrichment_ablation.py::TestEnrichmentAblation::test_enrichment_improves_rank_correlation PASSED
tests/regular_tests/test_exactly_once_processing.py::TestExactlyOnceProcessing::test_10k_duplicates_produce_single_row PASSED
tests/regular_tests/test_exactly_once_processing.py::TestExactlyOnceProcessing::test_process_event_level_idempotency PASSED
tests/regular_tests/test_integration_pipeline.py::test_db_agent_schema_creates_customer_risk_profile_table PASSED
tests/regular_tests/test_integration_pipeline.py::test_handle_customer_risk_profile_persists_correct_values PASSED
tests/regular_tests/test_integration_pipeline.py::test_handle_customer_risk_profile_is_idempotent PASSED
tests/regular_tests/test_integration_pipeline.py::test_event_type_contract_aggregator_to_db_agent
```

```
PASSED
tests/regular_tests/test_integration_pipeline.py::test_agent_supervisor_stores_launch_fn
PASSED
tests/regular_tests/test_integration_pipeline.py::test_agent_supervisor_restart_without_launch_fn_re
turns_false PASSED
tests/regular_tests/test_ml_payment_prediction.py::TestPaymentPredictionML::test_model_warmup
PASSED
tests/regular_tests/test_ml_payment_prediction.py::TestPaymentPredictionML::test_process_event_2_sta
ge_policy PASSED
tests/regular_tests/test_ml_payment_prediction.py::TestPaymentPredictionML::test_shap_values_integri
ty PASSED
tests/regular_tests/test_overpayment_clamping.py::TestOverpaymentClamping::test_overpayment_clamps_t
o_total PASSED
tests/regular_tests/test_overpayment_clamping.py::TestOverpaymentClamping::test_query_agent_never_re
turns_negative_remaining PASSED
tests/regular_tests/test_query_client_thread_isolation.py::TestQueryClientThreadIsolation::test_conc
urrent_queries_no_contamination PASSED
tests/regular_tests/test_query_client_thread_isolation.py::TestQueryClientThreadIsolation::test_thre
ading_local_provides_isolation PASSED
tests/regular_tests/test_query_client_thread_isolation.py::TestQueryClientThreadIsolation::test_wron
g_correlation_id_is_rejected PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_decisio
n_matrix_mapping[DEGRADED_high_lag] PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_decisio
n_matrix_mapping[CRITICAL_no_lag] PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_decisio
n_matrix_mapping[ERROR_low_cpu] PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_decisio
n_matrix_mapping[OVERLOADED_high_cpu] PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_decisio
n_matrix_mapping[TIMEOUT] PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_score_b
oundary_at_degraded_threshold PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_score_b
oundary_at_scale_threshold PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_score_b
oundary_at_critical_threshold PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_score_b
elow_degraded_threshold PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_score_a
t_maximum PASSED
tests/regular_tests/test_recovery_action_correctness.py::TestRecoveryActionCorrectness::test_score_a
t_zero PASSED
tests/regular_tests/test_recovery_storm_prevention.py::TestRecoveryStormPrevention::test_duplicate_d
egraded_events_suppressed PASSED
tests/regular_tests/test_risk_score_monotonicity.py::TestRiskScoreMonotonicity::test_avg_delay_monot
onic PASSED
tests/regular_tests/test_risk_score_monotonicity.py::TestRiskScoreMonotonicity::test_on_time_ratio_m
onotonic PASSED
tests/regular_tests/test_risk_score_monotonicity.py::TestRiskScoreMonotonicity::test_overdue_count_m
onotonic PASSED
tests/regular_tests/test_risk_score_monotonicity.py::TestRiskScoreMonotonicity::test_generate_monoto
nicity_chart PASSED
tests/regular_tests/test_self_healing_concurrent.py::TestSelfHealingConcurrent::test_rlock_no_deadlo
ck_10x10 PASSED
tests/regular_tests/test_throughput_scaling.py::TestThroughputScaling::test_throughput_scales_with_r
eplicas PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_customer_identity_contract
PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_metrics_enrichment_with_company_name
PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_consumer_group_scaling_canonical_group_id
PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_external_agent_fallback_chain
PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_producer_only_agent_lifecycle
PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_base_agent_start_skips_kafka_subscription_
for_producer_only_agent PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_lazy_kafka_producer_init
```

```
PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_runtime_manager_spawn_request_requests_single_placement_with_incremented_replica_count PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_placement_engine_preserves_restart_context PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_self_healing_emits_single_spawn_request_with_placement_hints PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_db_agent_preserves_existing_invoice_total_when_status_update_omits_amount PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_memory_agent_recompute_state_tolerates_null_invoice_amounts PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_customer_state_metrics_tolerate_null_invoice_amounts PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_query_agent_clamps_negative_remaining_amounts PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_lock_contention_optimization PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_self_healing_agent_bug_fix PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_db_agent_handles_payment_partial_with_string_amount PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_db_agent_rejects_non_numeric_payment_amount PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_kafka_client_init_consumer_rejects_unknown_backend PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_memory_agent_persist_metrics_creates_missing_customer PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_db_agent_repair_payment_integrity_backfills_orphans_and_clamps_paid PASSED
tests/regular_tests/test_unit_architecture_fixes.py::test_external_scraping_news_analysis_uses_description_keywords PASSED
tests/regular_tests/test_unit_circuit_breaker.py::test_circuit_breaker_failures_open_circuit PASSED
tests/regular_tests/test_unit_circuit_breaker.py::test_circuit_breaker_open_raises_immediately PASSED
tests/regular_tests/test_unit_circuit_breaker.py::test_circuit_breaker_recovery_to_half_open PASSED
tests/regular_tests/test_unit_circuit_breaker.py::test_circuit_breaker_half_open_to_closed PASSED
tests/regular_tests/test_unit_datetime_tz.py::TestEventTimezoneNormalisation::test_tz_aware_string_parsed_to_naive_datetime PASSED
tests/regular_tests/test_unit_datetime_tz.py::TestEventTimezoneNormalisation::test_tz_aware_string_is_a_datetime_instance PASSED
tests/regular_tests/test_unit_datetime_tz.py::TestEventTimezoneNormalisation::test_utc_string_no_offset_parsed_correctly PASSED
tests/regular_tests/test_unit_datetime_tz.py::TestEventTimezoneNormalisation::test_naive_utcnow_roundtrip PASSED
tests/regular_tests/test_unit_datetime_tz.py::TestEventTimezoneNormalisation::test_timezone_aware_utc_stripped PASSED
tests/regular_tests/test_unit_datetime_tz.py::TestEventTimezoneNormalisation::test_positive_offset_normalised PASSED
tests/regular_tests/test_unit_datetime_tz.py::TestEventTimezoneNormalisation::test_negative_offset_normalised PASSED
tests/regular_tests/test_unit_event_validation.py::test_valid_event_passes_validation PASSED
tests/regular_tests/test_unit_event_validation.py::test_wrong_schema_version_goes_to_dlq PASSED
tests/regular_tests/test_unit_event_validation.py::test_invalid_event_type_goes_to_dlq PASSED
tests/regular_tests/test_unit_event_validation.py::test_missing_required_field_goes_to_dlq PASSED
tests/regular_tests/test_unit_external_agent.py::TestExternalDataAgentNonBlocking::test_handle_event_returns_before_http_completes PASSED
tests/regular_tests/test_unit_external_agent.py::TestExternalDataAgentNonBlocking::test_handle_event_submits_to_executor_not_inline PASSED
tests/regular_tests/test_unit_external_agent.py::TestExternalDataAgentNonBlocking::test_handle_event_missing_customer_id_returns_early PASSED
tests/regular_tests/test_unit_external_agent.py::TestExternalDataAgentNonBlocking::test_handle_event_does_not_block_consumer_thread PASSED
tests/regular_tests/test_unit_kafka_client.py::TestKafkaClientCommitOffset::test_commit_with_message
```

```

_commits_offset_plus_one PASSED
tests/regular_tests/test_unit_kafka_client.py::TestKafkaClientCommitOffset::test_commit_with_message
_does_not_call_bare_commit PASSED
tests/regular_tests/test_unit_kafka_client.py::TestKafkaClientCommitOffset::test_commit_without_mess
age_uses_bare_commit PASSED
tests/regular_tests/test_unit_kafka_client.py::TestKafkaClientCommitOffset::test_commit_offset_is_ex
actly_message_offset_plus_one PASSED
tests/regular_tests/test_unit_kafka_client.py::TestKafkaClientCommitOffset::test_commit_only_touches
_correct_partition PASSED
tests/regular_tests/test_unit_kafka_client.py::TestKafkaClientCommitOffset::test_commit_when_no_cons
umer_is_a_no_op PASSED
tests/regular_tests/test_unit_kafka_client.py::TestKafkaClientCommitOffset::test_commit_propagates_c
onsumer_exceptions PASSED
tests/regular_tests/test_unit_placement.py::test_placement_routing_round_robin
PASSED
tests/regular_tests/test_unit_placement.py::test_placement_routing_no_capable_agent
PASSED
tests/regular_tests/test_unit_placement.py::test_placement_routing_registry_update
PASSED
tests/regular_tests/test_unit_query_client.py::TestQueryClientThreadIsolation::test_same_thread_reus
es_consumer PASSED
tests/regular_tests/test_unit_query_client.py::TestQueryClientThreadIsolation::test_different_thread
s_get_different_consumers PASSED
tests/regular_tests/test_unit_query_client.py::TestQueryClientThreadIsolation::test_concurrent_query
_calls_timeout_independently PASSED
tests/regular_tests/test_unit_query_client.py::TestQueryClientThreadIsolation::test_no_cross_contami
nation_of_correlation_ids PASSED
tests/regular_tests/test_unit_schema.py::test_schema_creation
PASSED
tests/regular_tests/test_unit_schema.py::test_customer_insertion
PASSED
tests/regular_tests/test_unit_schema.py::test_invoice_queries
PASSED
tests/regular_tests/test_unit_schema.py::test_get_customer_with_enrichment_fields
PASSED
tests/regular_tests/test_unit_self_healing.py::TestSelfHealingLockSafety::test_handle_degraded_compl
etes_without_deadlock PASSED
tests/regular_tests/test_unit_self_healing.py::TestSelfHealingLockSafety::test_publish_restart_reacq
uires_rlock_while_held PASSED
tests/regular_tests/test_unit_self_healing.py::TestSelfHealingLockSafety::test_evaluate_state_sets_l
ast_restart_requested PASSED
tests/regular_tests/test_unit_self_healing.py::TestSelfHealingLockSafety::test_handle_degraded_withi
n_grace_period_skips_restart PASSED
tests/regular_tests/test_unit_self_healing.py::TestSelfHealingLockSafety::test_concurrent_handle_deg
raded_does_not_deadlock PASSED
tests/regular_tests/test_unit_self_healing_health_routing.py::test_monitoring_agent_health_topic_is_
acis_agent_health PASSED
tests/regular_tests/test_unit_self_healing_health_routing.py::test_monitoring_agent_publishes_degrad
ed_to_health_topic PASSED
tests/regular_tests/test_unit_self_healing_health_routing.py::test_self_healing_agent_subscribes_to_
health_topic PASSED
tests/regular_tests/test_unit_self_healing_health_routing.py::test_self_healing_agent_subscribed_top
ics_init_matches_subscribe PASSED
tests/regular_tests/test_unit_self_healing_health_routing.py::test_self_healing_process_event_routes
_degraded_to_handle_degraded PASSED
tests/regular_tests/test_unit_self_healing_health_routing.py::test_self_healing_process_event_routes
_critical_to_handle_critical PASSED
tests/regular_tests/test_unit_self_healing_health_routing.py::test_self_healing_drops_stale_health_e
vents PASSED
tests/regular_tests/test_unit_self_healing_health_routing.py::test_handle_degraded_sets_last_degrade
d_at PASSED

```

Command 2 — `python -m pytest tests/suite/ -v`

Environment

```
platform darwin -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0
rootdir: /Users/johangeorge/ACIS-X
configfile: pytest.ini
plugins: anyio-4.14.2
collected 45 items
```

Individual Test List

```
tests/suite/test_dynamic_discovery.py::TestDynamicDiscovery::test_agent_dynamic_registration
PASSED [ 4%]
tests/suite/test_dynamic_discovery.py::TestDynamicDiscovery::test_multi_agent_concurrent_discovery
PASSED [ 8%]
tests/suite/test_integration_contracts.py::TestRiskProfilePersistence::test_persists_correct_values
PASSED [ 11%]
tests/suite/test_integration_contracts.py::TestRiskProfilePersistence::test_idempotent_double_write
PASSED [ 13%]
tests/suite/test_integration_contracts.py::TestRiskProfilePersistence::test_event_type_contract_aggr
egator_to_db_agent PASSED [ 15%]
tests/suite/test_integration_contracts.py::TestCSAtoPPAContract::test_metrics_event_contains_require
d_fields PASSED [ 17%]
tests/suite/test_integration_contracts.py::TestRSAtoCollectionsContract::test_risk_scored_contains_r
equired_fields PASSED [ 20%]
tests/suite/test_integration_contracts.py::TestCorrelationIdPropagation::test_correlation_id_flows_t
hrough_csa_ppa_rsa PASSED [ 22%]
tests/suite/test_intelligence_comparison.py::TestACISvsNaiveBaseline::test_acis_f1_exceeds_naive_f1
PASSED [ 24%]
tests/suite/test_intelligence_comparison.py::TestSignalFusionVsSingleSignals::test_acis_rank_correla
tion_dominates_single_signals PASSED [ 26%]
tests/suite/test_intelligence_comparison.py::TestExternalEnrichmentAblation::test_enrichment_improve
s_rank_correlation PASSED [ 28%]
tests/suite/test_lifecycle_trace.py::TestInvoiceFullLifecycle::test_single_invoice_lifecycle_complet
es PASSED [ 31%]
tests/suite/test_lifecycle_trace.py::TestInvoiceFullLifecycle::test_correlation_id_propagates_throug
h_pipeline PASSED [ 33%]
tests/suite/test_lifecycle_trace.py::TestInvoiceFullLifecycle::test_stage_latency_breakdown_N50
PASSED [ 35%]
tests/suite/test_lifecycle_trace.py::TestPaymentRescoreLifecycle::test_payment_triggers_risk_rescore
PASSED [ 37%]
tests/suite/test_performance.py::TestPipelineP95Latency::test_p95_latency_under_500ms
PASSED [ 40%]
tests/suite/test_performance.py::TestSelfHealingMTTR::test_mttr_50_iterations
PASSED [ 42%]
tests/suite/test_performance.py::TestCollectionsAgentDuplicatePrevention::test_no_duplicate_actions_
for_repeated_risk_scored PASSED [ 44%]
tests/suite/test_self_healing_proof.py::TestAgentHardFailure::test_hard_failure_triggers_restart
PASSED [ 46%]
tests/suite/test_self_healing_proof.py::TestLatencySpike::test_latency_spike_triggers_recovery
PASSED [ 48%]
tests/suite/test_self_healing_proof.py::TestCascadingFailureStormPrevention::test_each_failed_agent_
recovers_exactly_once PASSED [ 51%]
tests/suite/test_self_healing_proof.py::TestRecoverySuccessRate::test_100_cycles_success_rate
PASSED [ 53%]
tests/suite/test_self_healing_proof.py::TestRecoveryTimeDistribution::test_recovery_time_distributio
n PASSED [ 55%]
tests/suite/test_unit_core.py::TestDBAgentContracts::test_invoice_amount_preserved_on_status_update
PASSED [ 57%]
tests/suite/test_unit_core.py::TestDBAgentContracts::test_overpayment_clamps_paid_amount
PASSED [ 60%]
tests/suite/test_unit_core.py::TestDBAgentContracts::test_exactly_once_db_level_guard
PASSED [ 62%]
tests/suite/test_unit_core.py::TestDBAgentContracts::test_query_agent_clamps_negative_remaining
PASSED [ 64%]
tests/suite/test_unit_core.py::TestRiskScoreMonotonicity::test_avg_delay_monotonic
PASSED [ 66%]
tests/suite/test_unit_core.py::TestRiskScoreMonotonicity::test_on_time_ratio_monotonic
PASSED [ 68%]
```

```
tests/suite/test_unit_core.py::TestRiskScoreMonotonicity::test_overdue_count_monotonic
PASSED [ 71%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.1-None] PASSED [ 73%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.29-None] PASSED [ 75%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.3-send_reminder] PASSED [ 77%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.5-send_reminder] PASSED [ 79%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.6-escalate_invoice] PASSED [ 81%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.75-escalate_invoice] PASSED [ 82%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.8-hold_credit] PASSED [ 84%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.89-hold_credit] PASSED [ 86%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
0.9-legal_escalation] PASSED [ 88%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_base_severity_to_action[
1.0-legal_escalation] PASSED [ 90%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_overdue_count_escalates_
action PASSED [ 93%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_chronic_delay_escalates_
action PASSED [ 95%]
tests/suite/test_unit_core.py::TestCollectionsAgentDecisionThresholds::test_90_day_overdue_invoice_f
orces_critical_priority PASSED [ 97%]
tests/suite/test_unit_core.py::TestRiskScoringContextTTL::test_cleanup_evicts_expired_entries
PASSED [ 99%]
tests/suite/test_unit_core.py::TestRiskScoringContextTTL::test_cleanup_enforces_max_size
PASSED [100%]
```

Warnings

Both runs emit **DeprecationWarning: datetime.datetime.utcnow() is deprecated** across a large number of tests (~50 000+ warning instances total). These are **non-blocking** — all tests pass regardless. The warnings originate from calls to `datetime.datetime.utcnow()` in both production code and test helpers. The recommended migration is to replace these with `datetime.now(datetime.UTC)` for Python 3.11+.

Affected source files (production code):

File	Approximate warning count
agents/storage/db_agent.py	~10 000+
agents/intelligence/aggregator_agent.py	~500
agents/base/base_agent.py	~50
self_healing/core/self_healing_agent.py	~50

Affected test files:

File	Approximate warning count
tests/regular_tests/test_canonical_consumer_group_scaling.py	~600
tests/regular_tests/test_throughput_scaling.py	~1 000
tests/regular_tests/test_self_healing_concurrent.py	~100
tests/suite/test_self_healing_proof.py	~500
Various others	~50 each

Historical Context — Recent Fixes Reflected in These Results

The following bugs were diagnosed and fixed prior to this clean test run:

Fix	File(s) Changed	Symptom Before Fix
<code>OffsetAndMetadata</code> 3-arg constructor incompatible with <code>kafka-python</code> <code>2.2.3</code> on macOS	<code>runtime/kafka_client.py</code> , <code>tests/regular_tests/test_unit_kafka_client.py</code>	Flood of <code>OffsetAndMetadata.__new__()</code> takes 3 positional arguments but 4 were given errors; offset commits failing for every agent
Kafka broker rebalance storm at startup	<code>docker-compose.yml</code>	Continuous <code>Heartbeat failed for group ...</code> because it is rebalancing; caused by <code>KAFKA_GROUP_INITIAL_REBALANCE_DELAY_MS:</code> <code>0</code> with 20+ agents joining simultaneously
<code>KafkaClient</code> defaulting to <code>confluent</code> backend (not installed on this machine)	<code>tests/suite/test_dynamic_discovery.py</code>	<code>ModuleNotFoundError: No module named</code> <code>'confluent_kafka'</code> inside daemon threads; <code>_consumer</code> remained <code>None</code> ; assertion failure in dynamic discovery tests
<code>agent.agent_id</code> attribute does not exist on <code>BaseAgent</code>	<code>tests/suite/test_dynamic_discovery.py</code>	<code>AttributeError:</code> <code>'PaymentPredictionAgent' object has no</code> attribute <code>'agent_id'</code> ; corrected to <code>agent.instance_id</code>

Notes

- Tests marked `pytest.mark.integration` (in `test_dynamic_discovery.py`) require a live Kafka broker on `localhost:9092`. They auto-skip with `pytest.skip()` if no broker is reachable, and were run here with the Docker Compose cluster active.
- The `tests/suite/` run is a **strict subset** of `tests/` — all 45 suite tests appear in both run outputs.
- No tests are currently marked `xfail` or permanently skipped.